

Anushka Kataria

Software Developer and Machine Learning Enthusiast

Proficient in Data Structures and Algorithms

Passionate about solving complex problems through technology and critical thinking

Focused on backend development and building scalable server-side systems

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EDUCATION

Mody University of Science and Technology, Sikar, Rajasthan

Bachelor of Technology - Computer Science and Engineering; CGPA: 7.67

Aug 2023 - Present

Pragati Convent H.S. School, Ratlam, Madhya Pradesh

Intermediate (XII) - MP Board; Percentage: 88.2%

Jul 2021 - Apr 2022

EXPERIENCE

IPCA Laboratories

Machine Learning Intern

Jun 2026 - Jul 2026

- Built a text preprocessing and entity extraction pipeline in Python using rule-based NLP (spaCy) with a **custom pharma vocabulary**, tagging adverse drug reaction (ADR) entities across therapeutic categories (anti-hypertensives, anti-malarials, DMARDs).
- Trained a supervised binary classifier (Logistic Regression, scikit-learn) to triage ADR vs. non-ADR intake records, **achieving precision and recall above majority-class baseline** on held-out validation data.
- Implemented negation detection that **reduced false-positive flags**, correctly handling phrases like "no adverse reaction reported" in free-text intake entries.

TRAINING AND CERTIFICATIONS

IBM PBEL Virtual Internship

Machine Learning | FactRadars

Jun 2025 - Jul 2025

- Built a fake news classifier using TF-IDF and Logistic Regression/SVM, **evaluated on F1, ROC-AUC and PR-AUC** for class-imbalance robustness.
- Fine-tuned** a DistilBERT transformer over the classical baseline, **benchmarking accuracy-latency tradeoffs**.

PROJECTS

NexTrace - Real Time Log Analytics & Search Platform

Feb 2026 - Apr 2026

- Built a real-time log analytics platform with a custom **inverted-index search engine**, replacing linear log search with structured, sub-second multi-term queries.
- Implemented **posting-list construction** and **delta encoding** in C++ for efficient index compression and fast sorted-list intersection at scale.
- Designed a **hybrid storage layer** (SQL metadata + compressed binary posting lists) and a CLI supporting AND/OR queries and time-range filtering, benchmarked against 1M log entries.
- Tech Stack: C++, Python, SQL, Regex-based parsing, CLI

AsyncFlow - Distributed Task Queue & Worker Processing Platform

Sept 2025 - Nov 2025

- Designed a distributed task queue supporting **lease-based dequeue** and **visibility timeouts** to guarantee **at-least-once job delivery** across a multi-worker system.
- Built a Python worker pool with **heartbeat-based health monitoring**, **exponential backoff retries**, and a **dead-letter queue** for permanently failed jobs.
- Implemented **idempotency-key enforcement** to prevent duplicate job execution, backed by a PostgreSQL schema with composite indexing for efficient job scheduling.
- Tech Stack: C++, Python, PostgreSQL, REST API, OOP

SKILL SUMMARY

- Languages:** Python, C++
- Core CS:** Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Object-Oriented Programming (OOP), System Design Basics
- Machine Learning & Deep Learning:** Machine Learning, Deep Learning, NLP, scikit-learn, PyTorch, Transformers (DistilBERT)
- Backend & Databases:** FastAPI, SQL, PostgreSQL, SQLite
- Tools & Platforms:** Git, GitHub, MLflow, Docker
- Soft Skills:** Problem Solving, Critical Thinking, Communication, Adaptability, Time Management
- Currently Learning:** Generative AI

ACHIEVEMENTS

- AMD Slingshot Hackathon 2026** - Ranked in the top 100 teams nationwide and top 10 in the Jaipur region, out of 40,000+ participants across India, for TruthLens, a multimodal deepfake detection platform presented at the Regional Demo Day.